Repair Headroom, Not Dying Gates: When an Own-Segment Bias Helps Continual Learning — and When It Silently Reverses

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Abstract

A recent audit of dendritic context-gated protection (companion paper, S1) found that a one-hidden-layer active-dendrite network with winner-take-all routing fails to develop segment–task specialization, that the standard dominant-segment instrument is contaminated by dead neurons, and that a constant bias b on the own-segment dendritic input repairs the learning term ($\Delta\ell$ up to +4.6 at T=20, $t \geq 8$). We test whether these findings transfer beyond the bare-MLP family. **They do not — and the failure mode is informative.** On a convolutional active-dendrite network (4-layer conv trunk + GroupNorm + global average pooling feeding the identical dendritic gate; 160 runs, 10 seeds per cell), the repair collapses to noise ($\Delta\ell$ significant in 4/12 dose cells; largest effect +1.46 vs. the MLP's +0.8 to +4.6) and *final accuracy reverses* (Δavg down to -2.29, $t = -3.1$). The dead-neuron artifact disappears entirely — not because the instrument improved, but because conv+GroupNorm mortality is exactly zero where the bare MLP's was 57%. A decisive same-architecture control (GroupNorm inserted into the MLP's feature layer, 80 runs, the only delta) separates the two candidate explanations: mortality and repair-gain *decouple* (GroupNorm raises mortality to 76.5%/85.5% while collapsing the gain to 17%/26% of its bare-MLP value), so dying gates are not the mediating variable; instead, gain decays as the unbiased baseline strengthens, and on both strong-feature backbones the repair buys learning only while significantly worsening retention. We conclude with a practitioner-facing boundary: own-segment

bias repairs pay when the baseline is weak and its gates die; on strong-feature backbones they silently trade retention for learning, and the dominant-segment diagnostic must be paired with a mortality probe before it is trusted anywhere.

Keywords: continual learning, dendritic computation, active dendrites, reproducibility, negative results

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1. Introduction

Companion paper S1 [1] established, on a one-hidden-layer active-dendrite network (ADNet [2]) with k-winners-take-all routing, a three-link chain: (1) the network fails to develop segment–task specialization; (2) the standard dominant-segment instrument overstates this failure because dead neurons fall to segment 0 by argmax convention; (3) a constant bias b on the own-segment dendritic input causally repairs the learning term with a clean dose–response (22/22 dose cells positive, 20/22 significant at $T \leq 20$).

S1's own limitations section asks the question this paper answers: *does the chain transfer beyond the bare-MLP family?* The question matters because the repair is deliberately architecture-agnostic — a scalar added to one gate — and because the artifact mechanism (argmax over dead gates) should exist wherever segments and hard routing coexist [3, 4].

We test transfer with the same instrument stack and pre-registered criteria, on two new configurations:

- **ConvADNet** — a 4-layer convolutional trunk with GroupNorm and global average pooling feeding the *identical* dendritic gate, kWTA sparsifier, and protected-gradient scheme (160 runs; Sec. 3);
- **GN-MLP** — the original single-hidden-layer network with one GroupNorm layer inserted between the feature projection and the dendritic gate (80 runs; Sec. 4). This is a same-architecture control: every hyperparameter, the routing structure, and the protection mechanism are unchanged; the norm layer alone differs.

Both transfer attempts fail their pre-registered criteria, and the *pattern* of failure identifies what the repair actually needs. The conv network has exactly zero neuronal mortality — the artifact vanishes not because the instrument got better but because the pathology it measures is absent — and the repair collapses with it. The GroupNorm control then kills the natural "dying-gate" hypothesis: normalization *raises* mortality while *collapsing* the repair gain, so the two are decoupled. What tracks the gain instead is the strength of the unbiased baseline: where the baseline is strong (GroupNorm, convolution), there is little learning-term room to buy, and the price shifts entirely to retention — final accuracy reverses significantly on both strong-feature configurations (-3.88 and -6.55 on GN-MLP; -0.84 and -2.29 on ConvADNet).

Contributions..

1. A pre-registered, 160-run transfer test of the S1 repair to a convolutional active-dendrite network: the repair does not transfer (4/12 cells significant vs. 15/20 required) and final accuracy reverses (Sec. 3).
2. A zero-mortality finding: conv+GroupNorm keeps every gate alive, so the dominant-segment artifact of S1 is architecture-bound, not instrument-bound (Sec. 3).
3. A decisive same-architecture control (80 runs, one delta) showing mortality and repair-gain decouple: the "dying-gate" hypothesis is false, and baseline strength — not gate death — is the tracking variable (Sec. 4).
4. A practitioner boundary condition: own-segment bias repairs pay only on weak-baseline, high-mortality configurations; on strong-feature backbones they silently trade retention for learning (Sec. 6).

All experiments are pre-registered (criteria fixed before data collection), paired by seed, and reported with all cells — including those that failed.

2. Background: the S1 chain being tested

We summarise only what the transfer test requires; full protocol in S1 [1]. The model is an active-dendrite network [2]: features $t = \mathrm{ff}(x)$, a context-

conditioned gate $u = W_d\,\mathrm{ctx}$ reshaped to (n_{seg}, n_h), a winning segment per neuron $d = \max_{\mathrm{seg}} u$, hidden activity $h = \mathrm{kWTA}(t \odot \sigma(d))$ with k=64, and a linear readout. Gradients of the gate are protected: only the current task's segment trains. Protection is usage-localized EWC [5] on the feature and readout blocks (λ=1000); the protected-gradient and context-gating design follows [2, 6]. Tasks are Permuted-MNIST and Permuted-Fashion [7] at $T \in \{10, 20\}$.

The S1 findings under test:

1. **Anti-specialization.** The own-segment win rate stays at chance ($1/T$) throughout training.
2. **Dead-neuron artifact.** The naive win rate (all neurons) is inflated 0.62 vs. 0.11 (active) by argmax ties over dead gates; mortality at T=10 is 57.2%.
3. **Repair.** Adding $b \in \{1, 3, 10\}$ to the own-segment gate entry lifts the learning term $\Delta\ell = +0.8$ to $+4.6$ ($\Delta\ell > 0$ in 22/22 dose cells, significant in 20/22; paired $|t|$ up to 14.7), with final accuracy improving simultaneously in 19/22 cells.

Pre-registered criteria (fixed before any data).. G1 (repair generalizes): $\geq 15/20$ dose cells with paired $\Delta\ell > 0$ significant, $\geq 18/20$ sign-consistent, with a saturation guard ($\ell_0 > 90$ cells excluded). The pre-registration template carried S1's "20 cells" denominator over verbatim; the conv grid is 2 datasets $\times$ 2 depths $\times$ 3 doses = 12 cells (no saturated cells arose), and all criteria are evaluated against 12. G2 (artifact replicates): untrained-naive share exceeds active share by ≥ 0.05. E4 (dying-gate control): confirms if GN mortality ≤ 0.10 *and* the repair vanishes; refutes if mortality > 0.30 or gain retention $\geq 50\%$.

3. ConvADNet: the repair does not transfer

3.1. Architecture and equivalence of the instrument stack

ConvADNet replaces only the feature path of S1's ADNet:

$$x \mapsto [\text{Conv-GN-ReLU}] \times 4 \mapsto \text{GAP} \mapsto t \in \mathbb{R}^{2048},$$

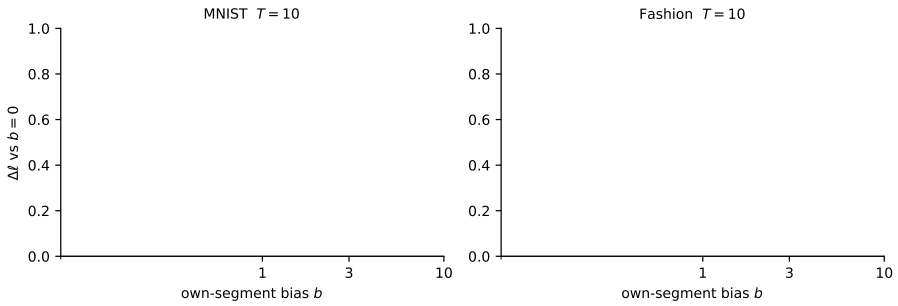


Figure 1: Dose–response of the own-segment bias across the three configurations (T=10, paired means vs. each configuration's own b=0). The bare MLP (blue) shows the S1 dose–response; GN-MLP (orange) collapses to a fifth of it; ConvADNet (green) to noise.

with strides $(1, 2, 2, 2)$ for 28×28 inputs. The dendritic gate, kWTA (k=64), readout, protected gradients, anchor construction (Fisher × usage), evaluation protocol, and the raw-gate routing instrument are *byte-identical* to S1: the gate reads dend(ctx) directly, so every routing statistic measures what the network encodes, not what a bias forces. Hyperparameters are unchanged (Adam 10^{-3}, 15 epochs, λ=1000, seeds 0–9 per cell). 160 runs total ($b \in \{0, 1, 3, 10\}$ × 2 datasets × $T \in \{10, 20\}$); zero failures.

3.2. G1 fails: the dose–response collapses

Table 1 reports paired differences against b=0. Against the pre-registered bar ($\geq$ 15/20 cells significant), the result is unambiguous: 4/12 **cells** significant, largest effect +1.46 (t=2.1), sign-consistency 11/12. In the five conv cells with a matched bare-MLP counterpart, the MLP's effects were 4–56× larger with paired $|t|$ up to 14.7 (Table 2, column $\Delta\ell$). Worse, final accuracy *reverses*: at fashion T=10, b=10 costs Δavg = −2.29 (t= − 3.1) — the repair buys a non-significant learning sliver at a significant accuracy cost.

3.3. G2 fails with a clean cause: zero mortality

The dominant-segment instrument reads identically zero artifact: naive share = active share = chance = $1/T$ in every cell (Table 2, column "dead"). The

Table 1: ConvADNet dose–response, paired vs. b=0 (n=10; *: $|t| > 2$). Pre-registered G1 requires $\geq 15/20$ cells with $\Delta\ell^*$; observed 4/12 (no saturated cells).

cell	arm (b)	$\Delta\ell$	Δavg	ΔBWT
MNIST T=10	1	+0.17	−1.64	−2.02
MNIST T=10	3	+0.30	−0.99	−1.43
MNIST T=10	10	+0.07	−0.84	−1.01
Fashion T=10	1	−0.06	+0.22	+0.31
Fashion T=10	3	+0.02	−0.89	−1.01
Fashion T=10	10	+0.36	−2.29*	−2.95*
MNIST T=20	1	+0.63	−0.24	−0.92
MNIST T=20	3	+1.46*	−0.26	−1.81
MNIST T=20	10	+1.28	+0.30	−1.04
Fashion T=20	1	+0.31*	+0.66	+0.36
Fashion T=20	3	+0.29*	−0.51	−0.84
Fashion T=20	10	+0.31*	−0.87	−1.24

cause is visible in the per-phase instrumentation: ConvADNet's dead-gate fraction is *exactly* 0.000 *in every phase of every run*, against the bare MLP's 0.572 [1]. GroupNorm's affine output keeps $|u| > 0$; there are no argmax ties over dead gates because there are no dead gates (Fig. 2).

A measurement caveat we state explicitly: the two configurations report mortality under *different* zero-conditions. The MLP instrument (inherited unchanged from S1) marks a neuron dead if its post-kWTA activity is never positive across the usage window — mortality there means "never wins the top-k competition". The conv instrument marks a neuron dead if its gate row is identically zero. Both are the natural operationalizations in their families, but they are not the same statistic, and the conv family's post-kWTA "never wins" rate was not measured. What the comparison supports is therefore the narrower claim we make: the S1 *artifact mechanism* (argmax ties over exactly-zero gates) has no substrate on the conv configuration — its precondition (numerically zero gates) never occurs. The artifact is thus **architecture-bound, not instrument-**

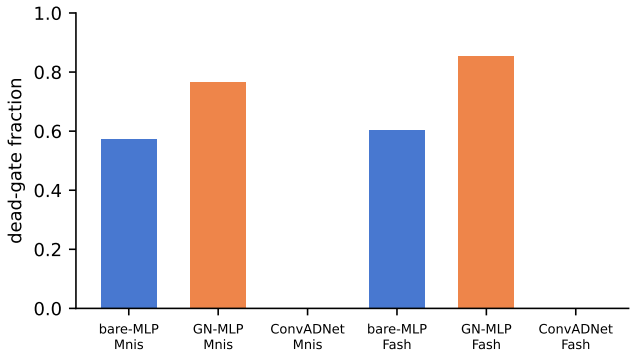


Figure 2: Dead-gate fraction at b=0 across configurations and datasets (T=10, mean over phases and seeds). Conv+GroupNorm mortality is exactly zero; GroupNorm inserted into the MLP *raises* mortality. Mortality is a property of the architecture, not of the intervention.

bound: the same reading instrument that S1 showed contaminated reads clean here, for reasons that have nothing to do with the instrument.

3.4. The conv baseline itself

ConvADNet's unbiased baseline learns more and forgets more than the MLP's: ℓ_0 84.1/75.3 (vs. 74.3/56.8) but BWT_0 $-50.4/-32.8$ (vs. $-11.6/-16.2$) at T=10. The protection strength λ=1000, calibrated for the MLP's parameter scale, does not carry over — itself a transfer finding we report rather than tune away (see Sec. 6).

4. GN-MLP: the dying-gate hypothesis, tested and rejected

The natural reading of Sec. 3 is: the repair worked in S1 because it revived dying gates, and conv gates do not die. This "dying-gate hypothesis" makes a sharp same-architecture prediction: insert a normalization layer into the MLP (features $t = \mathrm{GN}(\mathrm{ff}(x))$, the *only* delta; 80 runs, seeds 0–9, $b \in \{0, 1, 3, 10\}$, both datasets, T=10) — mortality should fall and the repair should weaken accordingly.

The prediction fails in both halves (Table 2, middle rows). Mortality *rises* (0.765/0.855 vs. bare's 0.572/0.603), for a reason the architecture makes plain:

Table 2: The three configurations at T=10 (n=10; paired vs. each configuration's own b=0). "dead" = mean dead-gate fraction; "HR_ℓ" = $100 - \ell_0$. The repair's $\Delta\ell$ tracks baseline weakness, not mortality.

dataset	config	ℓ_0	HR_ℓ	dead	$\Delta\ell_{b=10}$	$\Delta\mathrm{avg}_{b=10}$
MNIST	bare-MLP	74.3	25.7	0.572	$+3.32^*$	$+4.80^*$
MNIST	GN-MLP	93.5	6.5	**0.765**	$+0.57^*$	-3.88^*
MNIST	ConvADNet	84.1	15.9	**0.000**	$+0.07$	-0.84
Fashion	bare-MLP	56.8	43.2	0.603	$+1.39^*$	$+0.69$
Fashion	GN-MLP	79.3	20.7	**0.855**	$+0.36^*$	-6.55^*
Fashion	ConvADNet	75.3	24.7	**0.000**	$+0.36$	-2.29^*

GroupNorm rescales the features t symmetrically about zero, and the kWTA competition keeps the 64 *largest* values of $t \odot \sigma(d)$ — so every neuron whose normalized feature is predominantly negative can never enter the top-k and counts as dead under the usage instrument. GroupNorm does not touch the gate zero-set u; it manufactures post-kWTA death through the feature sign structure instead. Meanwhile the repair gain *collapses anyway*: to +0.57 (retention 17% of the bare-MLP gain) on MNIST and +0.36 (26%) on Fashion. And final accuracy reverses on both datasets (Δavg $= -3.88$, $t = -5.3$; -6.55, $t = -4.4$).

Mortality and gain are therefore **decoupled**: one configuration (GN-MLP) has the highest mortality and the least gain; another (ConvADNet) has zero mortality and no gain; the bare MLP has middle mortality and full gain. Whatever the repair feeds on, it is not gate death per se.

5. What actually tracks the gain

If not mortality, what? Table 2 orders the six cells by $\Delta\ell_{b=10}$: the bare MLPs lead on both datasets, the strong-baseline configurations trail, and within MNIST the ordering matches the unbiased baseline's learning-term weakness exactly (ℓ_0 74.3 $\to$ 84.1 $\to$ 93.5 versus gain +3.32 $\to$ +0.07 $\to$ +0.57; Fig. 3). We call this loose regularity *repair headroom*: the bias can only buy learning-

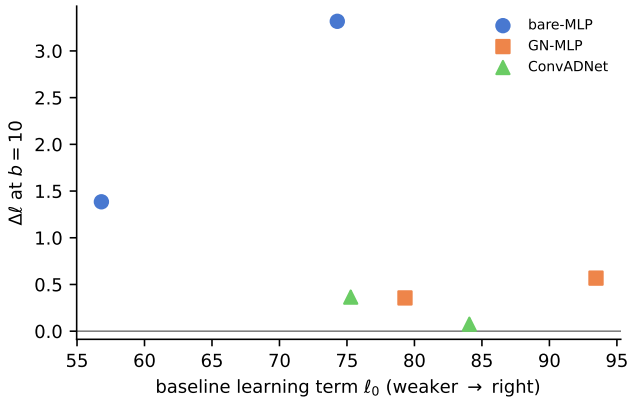


Figure 3: Repair gain at b=10 against the unbiased baseline's learning term. The qualitative regularity (weak baseline, larger gain) holds within MNIST and Fashion separately, but a quantitative law fails: ConvADNet-Fashion has 24.7 points of ℓ-headroom and no significant gain. Headroom is necessary-looking, not sufficient.

term room that the unbiased baseline leaves open.

We stress what the data do *not* support. A quantitative headroom law fails: across the six configuration×dataset cells the Pearson correlation between HR_ℓ and $\Delta\ell$ is $r = 0.38$ (cell means; pooled over per-seed differences, using each cell's mean HR_ℓ, $r = 0.27$; using each seed's own $100 - \ell_0$ gives $r = 0.31$), and the clearest counterexample is ConvADNet on Fashion — 24.7 points of ℓ-headroom, gain +0.36 (t=1.4, not significant). Headroom is thus **necessary-looking but not sufficient**: a strong baseline forecloses the repair (GN-MLP), but an open baseline does not guarantee it (convolution adds an independent obstacle — plausibly the routing prior interacting with distributed conv features, a question we leave open).

What is uniform across both strong-feature configurations is the *sign structure*: $\Delta\ell \geq 0$ (small) while $\Delta\mathrm{avg} < 0$ (significant in 3/4 cells; Fig. 4). The repair never destroys the learning term, but on strong backbones it buys that term by trading retention — exactly the trade S1's identity decomposition predicts, sharpened into a reversal once the learning side has no room left to absorb the cost.

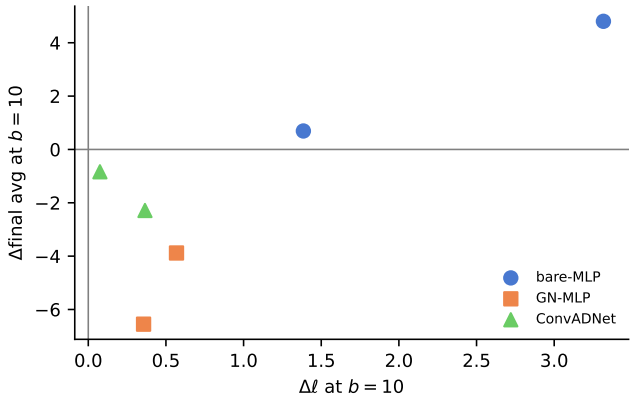


Figure 4: What the bias buys and what it charges (T=10, b=10, paired means). The bare MLP sits in the gain–gain quadrant; every strong-feature cell falls in the learning-vs-retention trade quadrant, with final accuracy significantly negative in 3/4.

6. Discussion and boundary conditions

For practitioners.. An own-segment bias is cheap, architecture-agnostic, and — on weak baselines with hard routing — effective [1]. This paper draws its boundary:

- **Audit mortality before trusting any dominant-segment statistic.** The S1 artifact appears iff gates die; conv+GN mortality is zero and the statistic reads clean. One probe (an untrained-pass share, or the dead-gate fraction) is enough to know which regime you are in.
- **Audit baseline strength before deploying the repair.** If the unbiased baseline's learning term is already near its ceiling, the bias will not buy learning — but will still charge retention ($\Delta\text{avg} \leq -2.3$ on every strong-feature cell, significant in 3/4).
- **Protection strength does not transfer.** λ calibrated on one parameter scale left the conv baseline with 2–4.4× the MLP's forgetting. Any cross-architecture comparison of protection methods must re-calibrate or report the uncalibrated gap explicitly.

For the S1 story.. The companion paper's chain survives intact on its home architecture, and this paper sharpens its mechanism claim: the repair was never a routing *prior* in the strong sense (S1's task-relevance probe already showed zero-relevance tasks are untouched); it is a structured capacity intervention whose raw material — weak-baseline, dying-gate networks — is architecture-dependent. "Structured capacity with an architecture-bound supply."

Threats to validity.. (i) The conv λ was not re-calibrated; a per-architecture sweep might restore some repair effect, though the GN-MLP control — where the MLP-calibrated λ *is* correct — shows the same collapse, so mis-calibration is not the whole story. (ii) One conv topology; the zero-mortality finding is a property of GroupNorm's affine output and should replicate across conv trunks, but transformer backbones with routing modules remain untested. (iii) Six configuration×dataset cells bound how strongly any cross-configuration regularity can be asserted; we deliberately report the failed quantitative version (Sec. 5) rather than promoting the qualitative one beyond its evidence.

7. Conclusion

Two pre-registered transfer attempts, 240 runs in total, both failed — and the failures triangulate the mechanism. The convolutionary active-dendrite network has no dead gates, no dominant-segment artifact, and no repair; the GroupNorm control has *more* dead gates and *less* repair. Gate death is neither necessary nor sufficient for the own-segment bias to work. What the repair needs is a weak baseline with learning-term room to buy; what it always charges is retention, and on strong-feature backbones — where nothing else is for sale — that is all it buys. Repairs of this family should be deployed with a mortality probe in one hand and a baseline audit in the other.

Data and code

All runs, pre-registrations (written before data collection), and analysis scripts are in the project repository: `https://github.com/zhouranxi/delp-anchor-budget`

(S3_CNN, S3b_E4 experiment series).

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